1 Lead Time Scheduling

Usage

You use the lead time scheduling function if you are set up to use HYDRA shop floor scheduling and your PPS/ ERP system does not allow for lead time scheduling to calculate operation-related basic dates.

Integration

Lead time scheduling calculates operation-related basic dates based on order-related basic dates – earliest start, earliest end, latest start, latest end. Operations are scheduled between these dates in HYDRA shop floor planning so as to assure that the order-related basic finish date is maintained.

Requirement

The following requirements must be met in order for orders to be scheduled by lead time scheduling:

- The order type can either be relevant for scheduling or relevant for detailed planning (customized setting).
- The order status can either be relevant for scheduling or relevant for detailed planning (customized setting).
- The operation status can either be relevant for scheduling or relevant for detailed planning (customized setting).
- The operation is flagged based on its processing code as either relevant for scheduling or relevant for detailed planning (customized setting).

This document will provide a description of which order types and status, or which processing code, are flagged as relevant for scheduling or relevant for detailed planning at the time of delivery.

- Scheduling is activated as a cyclic job in the scheduler (Script *hls_term.scr*).

1.1 Procedure

Lead time scheduling determines the start times and/or end times for all operations of an order and consequently also for the order itself. Lead time scheduling includes both forward and backward scheduling. In the operation, forward scheduling determines the earliest start (earliest start time/EST) and the earliest end (earliest end time/EET) while backwards scheduling determines the latest start (latest start time/LST) and the latest end (latest end time/LET).

Defining the EST/EET

In order to define EST/EET, you have the option to specify the production order start date; otherwise the current time will be set. However, the planning lead time defined in the HYDRA basic settings is also included in the calculation of the current time so that the earliest start date is "now" + the planning lead time. When calculating the dates, the system takes the so called factory calendar (i.e. the shift calendar that is designated as "factory calendar") into account.

Example

The following shifts are defined in the factory calendar: 5:45 a.m. - 8:00 a.m., 8:15 a.m. - 11:30 a.m., 12:00 noon - 1:45 pm, …; in the HYDRA basic settings 5 hours of planning lead time are set.

Start of scheduling: 10:20 a.m.

Calculation: 10:20 a.m. + 5 hours + 0:30 hours for shift break (11:30 a.m.-12:00 noon)

Results: The earliest start time for scheduling is therefore 3:50 pm

Defining the LST/LET

Either the production order finish date (customer deadline, delivery date) or the earliest end time of the production order is used to define the LST/LET.

Processing times

The total processing time is calculated for the individual operations for scheduling purposes. This total processing time is derived from the sum of the process times (durations) defined for the operations

- wait time
- setup time
- processing time/ remaining run time (see below)
- dismantling time/retooling (teardown) time
- idle time
- transport time

These process times are either set explicitly (manually or via interface) or calculated using defined formulas.

Processing time/ Remaining run time

The processing time/ remaining run time is calculated based on the remaining run time formula defined in the operation. In this context, the performance level defined for the workplace/ at the machine is taken into consideration. If the operation has not yet been planned in detail, then the performance level is used that is defined for the planned group.

During scheduling, the available capacity is considered, calculated based on the results of the shift model - year model field or otherwise the year model defined as deviating planning model at the planned workplace. If the operation is not yet assigned to any individual workplace, the group shift calendar is used. Exceptions to this are transition times that - depending on the transition time - are matched to the Gregorian calendar or even to a specific shift calendar (transport time).

Please pay attention to the fact that the default values that are used for calculating the remaining run time formula are edited/maintained completely/correctly at the operation, for new operations that are included in the pool of groups and that have not yet been planned.

Production variants are not taken into account for lead time scheduling.

Affected orders

Orders that match the following criteria are scheduled with lead time:

- The order is of the order type 0, 2, 3, 5 or 6.
- The order status is prepared or started.

If these order-related criteria are met, then in continuation the following operations are considered.

Affected operations

- The operation status is prepared, running, interrupted or automatically interrupted.
- The operation is flagged based on its processing code as either relevant for scheduling or relevant for detailed planning (customized setting).
  In the standard delivery, all operations receive the processing code SYSTEM. This is flagged as either relevant for scheduling or relevant for detailed planning.

General processing

The individual operations of the order are considered and started with scheduling, beginning with the first operation relevant for scheduling that is not yet finished. For operations already planned for workplaces, the planning dates are interpreted as scheduled dates (regardless of their scheduled dates), because it is assumed that the operation is processed by this date. If, on the other hand, the planning dates are set in the past, then they are ignored, i.e. the operation is rescheduled.

If the EET of an operation is later than the basic finish date of the production order, or the LST is earlier than the basic start date of the production order, then the basic finish date of the production order cannot be met. If a reduction strategy was assigned to the production order, then based on this strategy, a step-by-step attempt is now made to shorten the total duration of the order so that the basic finish date is not exceeded.

Each order is scheduled independently of the other orders. The scheduling result valid for the order (scheduled start time, scheduled end time) is defined based on the scheduling type in the order header. If no scheduling type is defined, then the scheduling type defined in the HYDRA basic settings is used.

If the order's basic start date is set in the future, then backwards scheduling sets the basic start date to "now + planning lead time". As a result, the basic start date of the order is ignored.

For operations that have the planning indicator at the processing code set to "N", processing time is treated as zero regardless of the processing times defined. In doing so, the start date of such operations is set to equal the scheduled finish date of the preceding operation and the finish date set to equal to the start date of the subsequent operation.

Planned operations are scheduled "as usual" (the planning dates are not taken into consideration). EST and LST are calculated. After scheduling is completed, the planning dates will overwrite the scheduled dates.

For running operations, the planned finish date is calculated and scheduling is continued from that time on.

Continued scheduling after OP end

If a finished operation is set in the past, the earliest start (in forward scheduling) of the successor operation is calculated based on the idle time, transport time and wait time. If the earliest start is > planning lead time, the earliest start of the successor is set to earliest start = now + planning lead time + wait time.

Results of scheduling

Results of lead time scheduling are updated dates for each operation.

The difference between LST and EST yields the buffer time for an operation. Operations with negative buffer times are deemed critical, because every delay to such an operation would result in a delay in the production order.

Calculating the order buffer or the delay when the buffer is negative:
This value is derived from the difference between the latest finish date (=requirement date) and the order's scheduled end time. Thus, the only way to make this buffer positive is by using forward scheduling. If the buffer is negative, then the order is late.

1.2 Reduction measures

It is possible to apply reduction measures as part of lead time scheduling:

If it turns out during scheduling that the lead time for a given order is longer than the allotted time available, then the system will attempt to take reduction measures to shorten the lead time accordingly.

Lead times are reduced in increments. To this end, the operation is repeated until either the deadline is met or the maximum reduction has been reached and it becomes inevitable that the deadline will be exceeded. In which steps each reduction takes place in the operation is defined by the reduction strategy that is referenced accordingly in the order header (Index tab *Dates*).

The reduction strategy can be applied to reduce the times listed below:

- Wait time, down to which, based on the steps of the reduction strategy, reductions can be made (at most: minimum value).
- Transport time, down to which, based on the steps of the reduction strategy, reductions can be made (at most: minimum value).

By scheduling using the reduction strategy described above you can reduce the lead time in order to meet the requirement date. Each result (reduction level) is recorded in the order header.

Please note:

The reduction strategies are defined during HYDRA customizing.

1.3 Executing lead time scheduling

The actions listed below cause an order to be scheduled. Keep in mind that only those orders are scheduled that meet the criteria described in operation flow:

- When an order or a separate operation is created this order will run through lead time scheduling. This applies equally to orders that are transferred via interface from the ERP/ PPS system as well as to those that were created manually.
- Planning, re-planning and deallocating (removing) operations in HYDRA shop floor scheduling. Keep in mind here that scheduling does not run until after the planning has saved.
- Modifications made to the operation manually using the operation editing functions at the client will cause the operation to be rescheduled or will initiate scheduling:
    - Modification of the workplace
    - Modification of the "external processing" flag
    - Modification of a target quantity
    - Modification of a target cycle

- Modification of process times (also by changes to parameters contained in formulas): Lead time, wait time, setup time, additional setup time, processing time, inspection time, dismantling/teardown time, idle time, synchronization time, transport time, delivery time

In all cases, the order number of the order to which a modification was made is stored in a table (*ade_auto_verarb*).

A process runs through each of the orders listed in the table cyclically. The periodicity in which this operation is started is set in the schedulerand it should be defined based on the system.

Scheduling can also be initiated manually. The relevant functions are described here.